ZENCUBE

ZAMIS V1 — Power BI Dashboard Build Guide

Step-by-step build instructions using the 18-domain candidate dataset

Scope: Phase 6 (Data Model) → Phase 7 (Five V1 Dashboards), built directly from the 18 domain CSV extracts derived from data_inventory_1000.csv.

1. Before You Open Power BI

This guide assumes Phase 1–3 (stakeholder discovery, data inventory, KPI catalogue) are already drafted, and picks up at Phase 6–7: building the analytical model and the five mandatory V1 dashboards from the 18 domain files.

1.1 Files you need

All 18 domain CSVs from 18_domain_datasets.zip, keyed on Candidate ID (1000 candidates) with one satellite table (13_Institution_CoE) keyed on Institution (12 institutions).

2. Step 1 — Build the Data Model

2.1 Import

- Power BI Desktop → Get Data → Text/CSV → import all 18 files.
- In Power Query, rename each query to match its domain (Student_Member, Assessment, Attendance_Engagement, … Faculty) — drop the numeric prefix.
- Set data types explicitly for every query (see Section 3 — do this before closing Power Query, not after).

2.2 Relationships

Set Student_Member as the hub table. All other Candidate-ID tables are 1:1 with it; Institution_CoE is 1:many outward to Student_Member on Institution.

From table	Key	Cardinality	Cross-filter
Assessment, Attendance_Engagement, Projects, Project_Registry_Artifacts, Sprint_Task, Rubrics_Reviews, Domain_Expert, Program_Manager, Talent_Registry, Mock_Interviews, Jobs_Placement, Commercial_Payments, Admissions_Enquiries, Platform_Usage, Challenge_Registry, Faculty	Candidate ID	1:1	Single, → Student_Member
Institution_CoE	Institution	1:many	Single, Institution_CoE → Student_Member

2.3 Data-type cleanup (do this in Power Query before loading)

Table	Columns	Action
Attendance_Engagement	Faculty_Attended, Faculty_Missed	Whole Number
Domain_Expert	Expert_Review_Count, Expert_Ready_Count	Whole Number
Program_Manager	Expert_Review_Count	Whole Number
Mock_Interviews	Mock_Interview_Count, Mock_Passed, Mock_Failed	Whole Number
Jobs_Placement	Job_Applications, Jobs_Shortlisted, Jobs_Interview_Scheduled, Jobs_Selected, Offers_Accepted	Whole Number
Admissions_Enquiries	Enquiry_Count, Enquiries_Converted, Enquiries_Followup, Enquiries_Lost	Whole Number
Challenge_Registry	Challenge_Count, Challenge_Best_Rank, Challenge_Winners	Whole Number

Table	Columns	Action
Faculty	Faculty_Activities, Faculty_Attended, Faculty_Missed, Faculty_Minutes	Whole Number
Student_Member	Registration Date	Date
Assessment	Initial Attempt Date, Sprint Attempt Date	Date
Attendance_Engagement, Platform_Usage	Last Login	Date
Talent_Registry	Last Profile Update	Date
Institution_CoE	Partner Since	Date

3. Step 2 — New Columns and Measures to Derive

The raw files only carry base facts. The KPIs the project guide requires (eligibility, at-risk, conversion funnels, trend deltas) must be built as calculated columns and DAX measures. Everything below is new — none of it exists in the source CSVs.

3.1 New calculated columns (add in the listed table)

#	Table	New Column	DAX (calculated column)
C1	Student_Member	Full Name	Full Name = Student_Member[First Name] & " " & Student_Member[Last Name]
C2	Student_Member	Program	Program = LEFT(Student_Member[Batch], FIND("-", Student_Member[Batch]) - 1)
C3	Student_Member	Batch Group (A/B)	Batch Group = RIGHT(Student_Member[Batch], 1)
C4	Assessment	Assessment Trend	Assessment Trend = IF(Assessment[Score Improvement %] > 0, "Improved", IF(Assessment[Score Improvement %] = 0, "No Change", "Declined"))
C5	Attendance_Engagement	Days Since Last Login	Days Since Last Login = DATEDIFF(Attendance_Engagement[Last Login], TODAY(), DAY)
C6	Attendance_Engagement	Attendance Band	Attendance Band = IF(Attendance_Engagement[Attendance %] >= 80, "Healthy", IF(Attendance_Engagement[Attendance %] >= 70, "Watch", "Low"))
C7	Rubrics_Reviews	Rubrics Score % (0–100)	Rubrics Score Pct = DIVIDE(Rubrics_Reviews[Rubrics_Avg_Score], 10) * 100 — rescales the 0–10 rubric scale to a 0–100 % used in KPI definitions
C8	Domain_Expert	Expert Readiness % (0–100)	Expert Readiness Pct = DIVIDE(Domain_Expert[Expert_Avg_Readiness], 10) * 100
C9	Sprint_Task	Sprint Quality % (0–100)	Sprint Quality Pct = DIVIDE(Sprint_Task[Sprint_Avg_Quality], 10) * 100
C10	Jobs_Placement	Funnel Stage Reached	Funnel Stage = SWITCH(TRUE(), Jobs_Placement[Offers_Accepted] > 0, "Joined/Offer Accepted", Jobs_Placement[Jobs_Selected] > 0, "Selected", Jobs_Placement[Jobs_Interview_Scheduled] > 0, "Interview Scheduled", Jobs_Placement[Jobs_Shortlisted] > 0,

#	Table	New Column	DAX (calculated column)
			"Shortlisted", Jobs_Placement[Job_Applications] > 0, "Applied", "No Applications")
C11	Commercial_Payments	Collection %	Collection Pct = DIVIDE(Commercial_Payments[Total_Paid], Commercial_Payments[Total_Invoiced])
C12	Commercial_Payments	Has Overdue Invoice	Has Overdue = IF(Commercial_Payments[Overdue_Invoices] > 0, "Yes", "No")
C13	Talent_Registry	Profile Ready Flag	Profile Ready = IF(AND(Talent_Registry[Talent Profile Status] = "Complete", Talent_Registry[Resume Score] >= 70), "Ready", "Not Ready")
C14	Institution_CoE	Institution Tenure (yrs)	Tenure Years = DATEDIFF(Institution_CoE[Partner Since], TODAY(), YEAR)

3.2 New DAX measures (add in a dedicated Measures table)

Create one blank table named _Measures (Modelling → New Table → _Measures = ROW("x",1)) and store every measure there so they aren't scattered across data tables.

Snapshot / core counts

#	Measure	DAX
M1	Snapshot Date	Snapshot Date = MAX(Attendance_Engagement[Last Login])
M2	Total Students	Total Students = DISTINCTCOUNT(Student_Member[Candidate ID])
M3	Active Students	Active Students = CALCULATE([Total Students], Attendance_Engagement[Attendance %] > 0)
M4	Activity Rate %	Activity Rate % = DIVIDE([Active Students], [Total Students])
M5	Avg Attendance %	Avg Attendance % = AVERAGE(Attendance_Engagement[Attendance %])
M6	Total Institutions	Total Institutions = DISTINCTCOUNT(Institution_CoE[Institution])

Assessment / Student Progress

#	Measure	DAX
M7	Avg Initial Score %	Avg Initial Score % = AVERAGE(Assessment[Initial Total %])
M8	Avg Sprint Score %	Avg Sprint Score % = AVERAGE(Assessment[Sprint Total %])
M9	Avg Score Improvement %	Avg Score Improvement % = AVERAGE(Assessment[Score Improvement %])
M10	% Students Improved	% Students Improved = DIVIDE(CALCULATE([Total Students], Assessment[Score Improvement %] > 0), [Total Students])
M11	Avg Rubric Score %	Avg Rubric Score % = AVERAGE(Rubrics_Reviews[Rubrics Score Pct])
M12	Avg Project Completion %	Avg Project Completion % = AVERAGE(Projects[Project_Avg_Completion])

Project Performance

#	Measure	DAX
M13	Projects Submitted (Total)	Projects Submitted = SUM(Projects[Project_Submitted])
M14	Projects Completed (Total)	Projects Completed = SUM(Projects[Project_Completed])

#	Measure	DAX
M15	Project Completion % (Org)	Project Completion % Org = DIVIDE([Projects Completed (Total)], [Projects Submitted (Total)])
M16	Delayed Projects (Sprint-Blocked)	Delayed Projects = CALCULATE([Total Students], Sprint_Task[Sprint_Blocked] > 0)
M17	Avg Sprint Task Completion %	Avg Sprint Completion % = AVERAGE(Sprint_Task[Task_Completion_Rate_Pct])
M18	Avg Sprint Quality %	Avg Sprint Quality % = AVERAGE(Sprint_Task[Sprint Quality Pct])

Talent Registry / Employability / Interviews

#	Measure	DAX
M19	Talent Registry Eligible	Talent Registry Eligible = VAR AttOk = CALCULATE([Total Students], Attendance_Engagement[Attendance %] >= 80) VAR RubOk = CALCULATE([Total Students], Rubrics_Reviews[Rubrics Score Pct] >= 80) VAR ProjOk = CALCULATE([Total Students], Projects[Project_Completed] > 0) RETURN CALCULATE([Total Students], Attendance_Engagement[Attendance %] >= 80, Rubrics_Reviews[Rubrics Score Pct] >= 80, Projects[Project_Completed] > 0)
M20	Talent Registry Eligible %	Talent Registry Eligible % = DIVIDE([Talent Registry Eligible], [Total Students])
M21	Mock Interview Pass Rate %	Mock Pass Rate % = DIVIDE(SUM(Mock_Interviews[Mock_Passed]), SUM(Mock_Interviews[Mock_Interview_Count]))
M22	Avg Mock Score	Avg Mock Score = AVERAGE(Mock_Interviews[Mock_Avg_Score])
M23	Interview-Ready Count	Interview Ready Count = CALCULATE([Total Students], Mock_Interviews[Mock_Avg_Score] >= 60)

Placement Funnel

#	Measure	DAX
M24	Applications	Applications = SUM(Jobs_Placement[Job_Applications])
M25	Shortlisted	Shortlisted = SUM(Jobs_Placement[Jobs_Shortlisted])
M26	Interviews Scheduled	Interviews Scheduled = SUM(Jobs_Placement[Jobs_Interview_Scheduled])
M27	Offers	Offers Selected = SUM(Jobs_Placement[Jobs_Selected])
M28	Offers Accepted (Joined)	Offers Accepted = SUM(Jobs_Placement[Offers_Accepted])
M29	Application → Offer Conversion %	App to Offer % = DIVIDE([Offers Accepted (Joined)], [Applications])
M30	Placed Students	Placed Students = CALCULATE([Total Students], Jobs_Placement[Placement_Status] IN {"Selected","Offer Accepted"})
M31	Placement % (of Eligible)	Placement % = DIVIDE([Placed Students], [Talent Registry Eligible])
M32	Avg CTC (LPA)	Avg CTC = AVERAGE(Jobs_Placement[Avg_CTC_LPA])
M33	Max CTC (LPA)	Max CTC = MAX(Jobs_Placement[Max_CTC_LPA])

At-Risk Students

#	Measure / column	DAX
C15 (col, Student_Member)	Risk Flag	Risk Flag = VAR Att = RELATED(Attendance_Engagement[Attendance %]) VAR DaysIdle = RELATED(Attendance_Engagement[Days Since Last Login]) VAR Rub = RELATED(Rubrics_Reviews[Rubrics Score Pct]) VAR Blocked =

#	Measure / column	DAX
		RELATED(Sprint_Task[Sprint_Blocked]) VAR RiskPoints = (IF(Att < 70, 1, 0)) + (IF(DaysIdle > 10, 1, 0)) + (IF(Rub < 60, 1, 0)) + (IF(Blocked > 0, 1, 0)) RETURN SWITCH(TRUE(), RiskPoints >= 3, "Red", RiskPoints = 2, "Amber", RiskPoints = 1, "Amber", "Green")
M34	Red-Risk Students	Red Risk Students = CALCULATE([Total Students], Student_Member[Risk Flag] = "Red")
M35	Amber-Risk Students	Amber Risk Students = CALCULATE([Total Students], Student_Member[Risk Flag] = "Amber")
M36	Inactive >10 Days	Inactive >10 Days = CALCULATE([Total Students], Attendance_Engagement[Days Since Last Login] > 10)
M37	Overdue Work (Sprint-Blocked)	Overdue Work Students = CALCULATE([Total Students], Sprint_Task[Sprint_Blocked] > 0)

Institutions / Revenue (supports CEO Overview)

#	Measure	DAX
M38	Total Invoiced	Total Invoiced = SUM(Commercial_Payments[Total_Invoiced])
M39	Total Collected	Total Collected = SUM(Commercial_Payments[Total_Paid])
M40	Outstanding Amount	Outstanding Amount = [Total Invoiced] - [Total Collected]
M41	Collection Rate %	Collection Rate % = DIVIDE([Total Collected], [Total Invoiced])
M42	Overdue Invoices (Count)	Overdue Invoices = SUM(Commercial_Payments[Overdue_Invoices])
M43	Admission Conversion %	Admission Conversion % = DIVIDE(SUM(Admissions_Enquiries[Enquiries_Converted]), SUM(Admissions_Enquiries[Enquiry_Count]))

4. Step 3 — Build the Five V1 Dashboards

Build in this order: CEO Overview and Student 360 first (Week 6 of the plan), then Student Progress, Project Performance and At-Risk (Week 7). Add a report-level date-refresh label and global slicers (Institution, Program, Batch) on every page's canvas or a synced-slicer panel.

4.1 Dashboard 1 — CEO Overview

Executive snapshot for Founder/Leadership: overall health of students, institutions, projects, employability and placement.

Tables/measures used

- Student_Member, Institution_CoE (filters)
- M2 Total Students, M3 Active Students, M4 Activity Rate %
- M6 Total Institutions
- M15 Project Completion % Org
- M34+M35 Red/Amber Risk counts
- M20 Talent Registry Eligible %
- M26 Interviews Scheduled, M28 Offers Accepted
- M31 Placement %, M39 Total Collected, M41 Collection Rate %

Filters/slicers on this page

Institution · Program · Batch · Snapshot Date (M1) shown as a card

Visuals to build

- 1. KPI card strip: Total Students, Active Students, Activity Rate %, Total Institutions.
- 2. KPI card strip 2: Talent Registry Eligible %, Interviews Scheduled, Offers Accepted, Placement %.
- 3. Column chart: Students by Institution, with Placement % as a secondary line (combo chart).
- 4. Donut: Student_Member[Risk Flag] distribution (Green/Amber/Red) — links to At-Risk dashboard via drillthrough.
- 5. Line chart: Avg Score Improvement % and Avg Attendance % across Batches (trend proxy, since no weekly timestamp exists).
- 6. Table: Placement funnel totals — Applications → Shortlisted → Interview → Offers → Joined (M24–M28).
- 7. Card: Collection Rate % and Outstanding Amount (M40) for a one-line revenue signal.

4.2 Dashboard 2 — Student 360

Single-candidate drill-through page for Program Manager / Domain Expert / Talent team to see one student's full journey.

Tables/measures used

- Student_Member (profile)
- Assessment
- Attendance_Engagement
- Projects, Sprint_Task, Rubrics_Reviews
- Talent_Registry
- Mock_Interviews
- Jobs_Placement

Filters/slicers on this page

Set as a Drillthrough page filtered by Candidate ID (drill from any visual showing a student)

Visuals to build

- 1. Header card group: Full Name, Institution, Department, Batch, Career Interest, Placement_Status.
- 2. Card row: Attendance %, Days Since Last Login, Risk Flag (color-coded).
- 3. Assessment mini-table: Initial vs Sprint (Aptitude/Technical/Coding/Total %) with Score Improvement % and Assessment Trend arrow.
- 4. Project & Sprint card group: Project_Count, Project_Completion_Rate_Pct, Sprint_Task_Count, Task_Completion_Rate_Pct, Sprint_Avg_Quality.
- 5. Rubrics & Domain Expert: Rubrics_Avg_Score, Rubrics Score %, Expert_Avg_Technical, Expert Readiness %.
- 6. Employability card group: Talent Profile Status, Profile Ready flag, Resume Score, Mock_Avg_Score, Mock_Pass_Rate_Pct.
- 7. Placement journey: a horizontal funnel/stepper from Job_Applications → Jobs_Shortlisted → Jobs_Interview_Scheduled → Jobs_Selected → Offers_Accepted using Funnel Stage Reached (C10).

4.3 Dashboard 3 — Student Progress

Cohort-level improvement view for Program/Academic teams: are students getting better over the program?

Tables/measures used

- Assessment (M7–M10)
- Attendance_Engagement (M5)
- Rubrics_Reviews (M11)
- Projects (M12)
- Student_Member (Batch, Institution filters)

Filters/slicers on this page

Institution · Program · Batch · Career Interest

Visuals to build

- 1. Clustered bar: Avg Initial Score % vs Avg Sprint Score % by Batch.
- 2. KPI card: Avg Score Improvement % (M9) with % Students Improved (M10) as a secondary label.
- 3. Scatter plot: Attendance % (x) vs Rubrics Score % (y), sized by Project_Avg_Completion, coloured by Performance Band — surfaces the engagement-vs-quality relationship.
- 4. Line/area chart: Avg Rubric Score % and Avg Project Completion % across Batch cohorts (cohort-order proxy for a time trend, since per-week history isn't in the extract).
- 5. Table: Batch-level rollup — Avg Attendance %, Avg Score Improvement %, Avg Rubric Score %, Talent Registry Eligible % — one row per Batch.
- 6. Gauge or KPI: Talent Registry Eligible % (M20) against a target (e.g. 40%).

4.4 Dashboard 4 — Project Performance

Delivery view for Program/DE teams: are projects and sprints progressing, and where are the blockers?

Tables/measures used

- Projects
- Project_Registry_Artifacts
- Sprint_Task
- Rubrics_Reviews

Filters/slicers on this page

Institution · Program · Batch

Visuals to build

- 1. KPI cards: Projects Submitted (M13), Projects Completed (M14), Project Completion % Org (M15), Delayed Projects (M16).
- 2. Bar chart: Project_Completion_Rate_Pct distribution banded (0–33/34–66/67–100) by Institution.
- 3. Column chart: Avg Sprint Task Completion % and Avg Sprint Quality % by Batch (M17, M18).
- 4. Stacked bar: Sprint_Completed vs Sprint_Blocked count per Institution — highlights concentration of blockers (Section 17's '8 delayed projects share the same dependency' example).
- 5. Table: Rubrics_Review_Count, Rubrics_Completed, Rubrics_Pending, Rubrics_Avg_Score by Batch — reviewer backlog visibility.

- 6. Scatter: Project_Avg_Score vs Sprint_Avg_Quality per student, coloured by Sprint_Blocked>0 flag.

4.5 Dashboard 5 — At-Risk Students

Actionable worklist for the Program Manager: who needs intervention today, and why.

Tables/measures used

- Student_Member (Risk Flag, C15)
- Attendance_Engagement
- Rubrics_Reviews
- Sprint_Task
- M34–M37

Filters/slicers on this page

Institution · Program · Batch · Risk Flag (Green/Amber/Red)

Visuals to build

- 1. KPI card strip: Red Risk Students (M34), Amber Risk Students (M35), Inactive >10 Days (M36), Overdue Work Students (M37).
- 2. Table (the core worklist): Candidate ID, Full Name, Institution, Batch, Attendance %, Days Since Last Login, Rubrics Score %, Sprint_Blocked, Risk Flag — sorted Red first.
- 3. Stacked bar: Risk Flag distribution by Institution / Batch, to spot concentration.